ExxonMobil

ExxonMobil - Quantitative Modelling Assignment

American option valuation and natural gas storage



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Introduction and overall results

Thank you for a carefully constructed assignment. I enjoyed revisiting these foundational papers and applying them to a practical storage problem. I built the four exercises as successive extensions of one least squares Monte Carlo (LSMC) framework, ending with daily storage decisions and an interactive tool that makes them visible.

Overall results

The table links each requested exercise to its main result and practical conclusion. RMSE denotes root mean square error.

Ex.	Assignment answered	Result and practical conclusion
1	LSMC; contract inputs, Euler steps and stochastic volatility.	Published prices reproduced with a mean absolute relative gap of 0.34%. Contract sensitivities have the expected directions; long-run variance is the main value driver in the tested setup.
2	European control; Barone-Adesi and Whaley; gamma.	Price-estimator variance falls 75–78%. A larger spot bump improves the tested 20% at-the-money gamma estimate: variance falls 7.8 times.
3	Milstein; pricing errors and convergence.	Milstein achieves strong order 1 versus Euler's 0.5 under GBM. Comparing price accuracy and runtime identifies when the extra simulation cost is useful.
4	Gas prices, feasible trades and storage value.	High-volatility value: EUR 4.5m; fitted decisions are 0.04% below our independent reference. The finer grid adds no material value. Sobol with a bridge cuts pricing RMSE by 74% versus antithetic Monte Carlo at 16,384 paths.

One project built progressively

Market inputs, contracts, simulation and valuation have separate responsibilities. Each exercise reuses the same foundation; Euler and Milstein feed the same option exercise algorithm. Storage adds inventory states and multiple feasible trades. Vectorised calculations and one shared regression factorisation per date handle that complexity. The browser tool reads saved results and makes prices, decisions and inventory directly inspectable.

Where the implementation required judgment

We retain the specified zero rates and derive the Barone-Adesi and Whaley limit. Exact GBM provides the European control and its known mean. The joint Milstein scheme requires cross iterated Itô integrals, simulated using Wiktorsson's series with a controlled truncation error. No matching full-year storage benchmark is supplied, so we build and refine an independent reference, separating decision quality from simulation noise.

Explore the tool in your browser

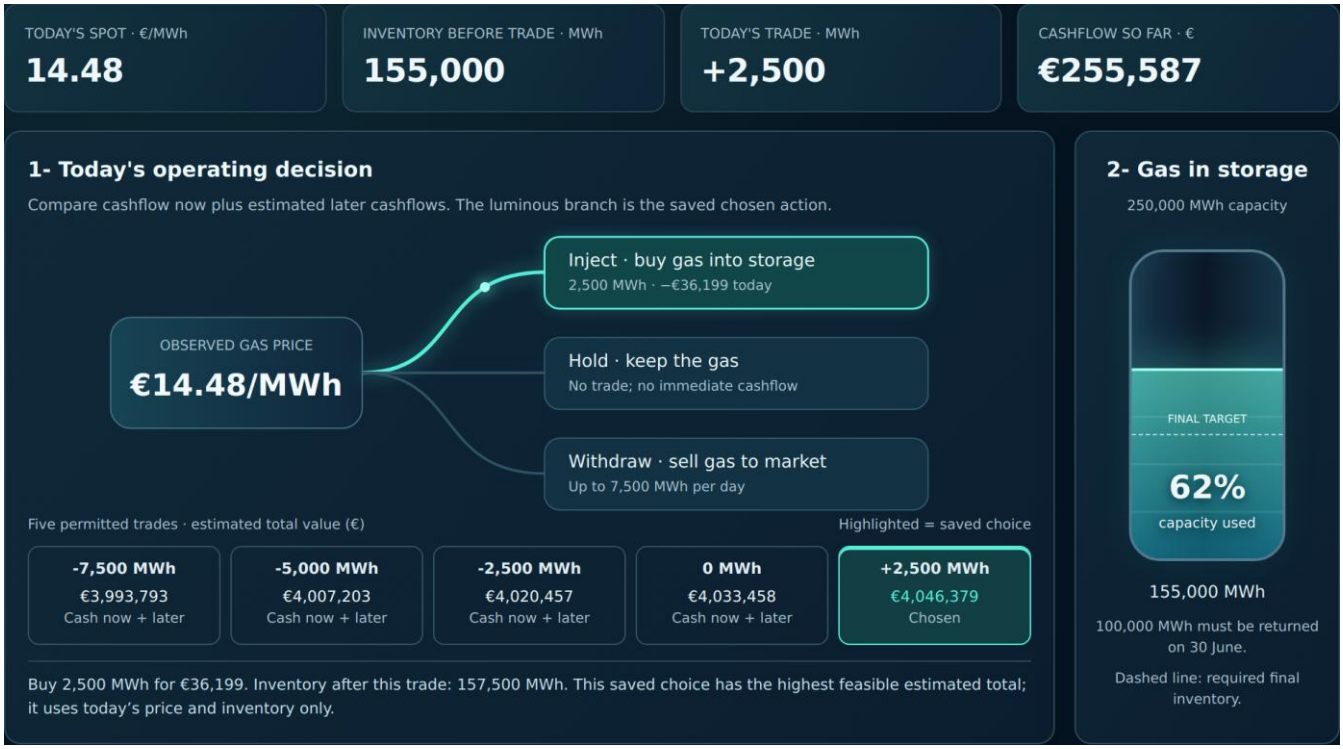
Extract the ZIP and follow README.txt's Python 3.12 installation and launch commands. Open http://localhost:8501: Project overview contains the replay; the sidebar opens each exercise.

The README also records installation alternatives. Starred controls replay the finer-grid and Sobol plus bridge studies; pages 25 and 26 explain their measured benefits.

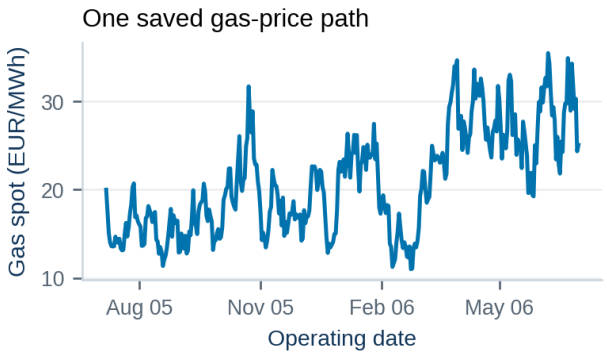
The gas storage tool

I built the gas storage tool to show the dynamic programme in action. For each day, it compares immediate trading cashflow plus estimated future value, selects a feasible trade, and updates inventory. The annual operating path connects these individual decisions to the full contract.

The tool reveals the decision: compare trading cashflow and future value, follow the selected action, and watch inventory change.



High-volatility sample path 1, 4 November 2005. Buy 2,500 MWh at EUR 14.48/MWh; inventory rises from 155,000 to 157,500 MWh. The cylinder shows inventory before the trade. The annual graphs below follow the same saved path.





Price is observed before each trade; inventory is shown after it. The price path creates trading opportunities, while capacity, daily flow limits and the final inventory requirement constrain the decisions.

This is one realised operating path. The contract valuation averages cashflows over independent scenarios: about EUR 4.5 million in the high-volatility case. A higher cashflow in another replay is a different outcome, not evidence of a higher expected contract value.

The replay opens on Project overview. Its controls explore saved baseline, finer-grid and Sobol plus bridge cases. Playback starts at four days per second. Launch guidance is on page 2.

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1.1 - American put valuation by regression

Continuation, exercise decisions and all 20 published prices

Our regression-based pricer reproduces the 20 published American put prices with a mean absolute relative gap of 0.34%.

An American put holder compares exercise with the expected discounted payment from waiting. For spot S_t, strike K, rate r and a later exercise date τ, define payment Y_t and continuation C_t:

$$Y_t = e^{-r(\tau - t)} \max(K - S_\tau, 0), \qquad C_t = \mathbb{E}[Y_t | \mathcal{F}_t].$$

$\mathcal{F}_t$ is information available at time t. For square-integrable Y_t, conditional expectation is its closest mean-square approximation using that information. Following Longstaff and Schwartz (2001), we regress on basis functions: X_t contains their values on in-the-money paths; β_t contains fitted coefficients.

$$\hat{\beta}_t = \operatorname{argmin}_\beta \| Y_t - X_t \beta \|^2, \qquad \hat{C}_t(s) = \phi(s)^\top \hat{\beta}_t.$$

Our basis ϕ = (1, x, x^2), x = S/K, captures level, slope and curvature with three coefficients. Strike scaling makes contracts comparable. We fit where S < K, since exercise pays only there; all paths enter valuation.

$$S_{n+1} = S_n + \mu S_n \Delta t + \sigma S_n \sqrt{\Delta t}\, Z_n, \qquad Z_n \sim N(0,1).$$

This Euler–Maruyama update simulates geometric Brownian motion (GBM): μ = r − q, σ is volatility, Δt the step in years and Z_n an independent standard normal shock. Published cases use r = 6% and continuous yield q = 0.

a) Learn decisions on 50,000 training paths. From maturity, work backwards: discount retained later payments and regress on the current state. NumPy least squares fits coefficients without a matrix inverse.

b) Exercise when max(K − S_t, 0) exceeds fitted continuation, retaining the actual payment. Apply the same coefficients, including the time-zero decision, on 100,000 new evaluation paths. Averaging discounted exercise payments measures the rule on independent scenarios; more evaluation paths improve precision.

$$\hat{V}_0 = \frac{1}{M} \sum_{i=1}^{M} D_i, \qquad D_i = e^{-r\tau_i} \max\left(K - S_{\tau_i}, 0\right), \qquad SE = \frac{s_D}{\sqrt{M}}.$$

M is the evaluation count, sD the payment sample standard deviation and SE the price standard error. SEs measure sampling uncertainty with fitted rules fixed. We also reproduce the article's eight-path algorithm check: value 0.114434.

Table 1.1. Our price minus published LSMC price (price units)

σ / years	S_0 = 36	S_0 = 38	S_0 = 40	S_0 = 42	S_0 = 44
20% / 1	-0.0121	+0.0011	-0.0030	-0.0164	-0.0114
40% / 1	-0.0103	-0.0187	-0.0200	-0.0241	-0.0211
20% / 2	-0.0015	-0.0083	-0.0072	-0.0175	+0.0012
40% / 2	+0.0189	-0.0180	+0.0039	+0.0172	-0.0185

Inputs are K = 40, days/365, at most one-day Euler steps and 50 exercise dates per year. The mean absolute gap is 0.0125 price units (0.34%); the maximum is 0.0241. All 20 differences lie within two combined standard errors from our and the article's estimates: close agreement across the published cases.

1.2 - Contract inputs and Euler steps

Strike, volatility and maturity tests; daily-to-weekly accuracy

Prices respond as expected to strike, volatility and maturity. Daily and weekly steps give similar accuracy in these tests.

We vary strike, volatility and maturity, holding $S_0 = 40$, r = 6%, zero yield and a one-day maximum Euler step fixed. Prices use the underlying's units.

Table 1.2. American put prices by strike, volatility and maturity (price units).

Volatility	Years	K = 36	K = 40	K = 44
20%	1	0.9122	2.3058	4.6501
20%	2	1.4313	2.8736	5.0605
40%	1	3.4154	5.2889	7.5760
40%	2	4.9790	6.9030	9.1958

Higher strike raises the payoff, higher volatility increases downside protection, and longer maturity adds exercise opportunities: all three price directions appear in the table. The four K = 40 cases also match published prices closely (mean absolute relative gap 0.28%).

Hold the 50 exercise dates per year fixed

To isolate simulation step size from exercise frequency, retain 50 exercise dates per year and repeat all 20 published contracts at maximum steps of 1, 2, 4 and 7 days. Figure 1.1 compares these 80 prices with Longstaff and Schwartz's (2001) finite-difference reference.

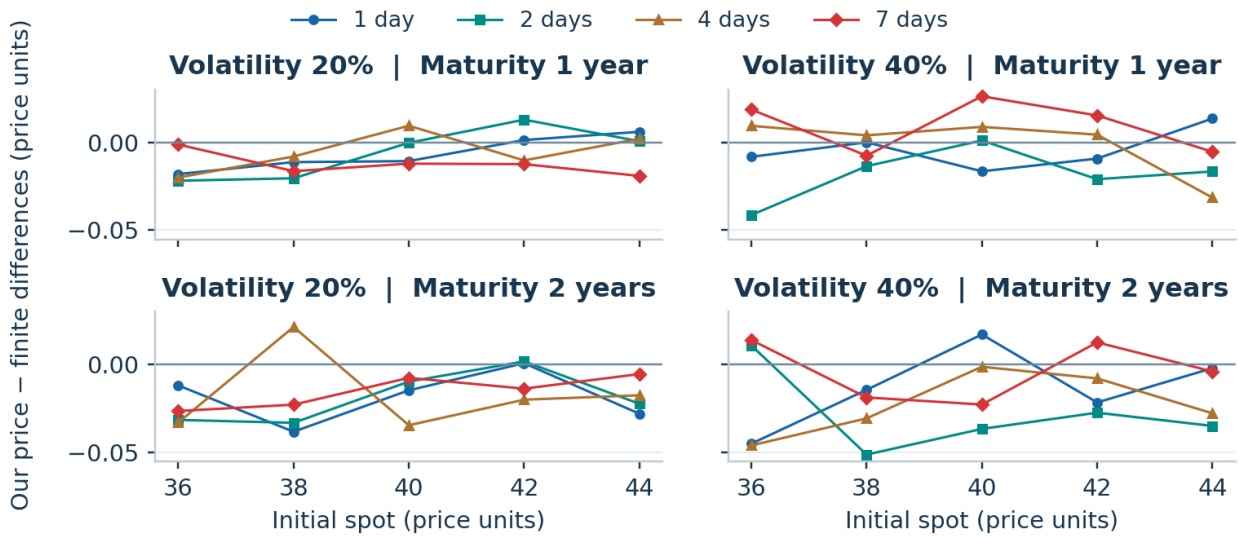


Figure 1.1. Estimated price minus finite-difference reference versus initial spot (price units). Panels separate volatility and maturity; colours distinguish maximum Euler steps. Zero indicates agreement.

Table 1.3. Mean absolute price gap to the published finite-difference reference (price units).

Mean absolute gap	1 day	2 days	4 days	7 days
σ = 20%	0.0141	0.0155	0.0177	0.0138
σ = 40%	0.0148	0.0255	0.0172	0.0145

Each cell averages ten contracts. Daily and weekly reference gaps are similar; intermediate steps show no consistent gain. The largest daily-to-weekly change is 0.0589; 19 of 20 changes lie within two combined evaluation SEs. Euler's omitted diffusion correction grows with $\sigma^2\Delta t$, but sampling and regression variation mask that trend here. Exercise 3 tests the correction directly.

1.3 - Repeat under stochastic volatility

The printed model, two-state regression and numerical studies

We extend the same valuation algorithm to spot and variance as two state variables, then repeat the contract and timestep tests.

$$dS_t=\mu S_t\,dt+\sqrt{v_t}S_t\,dW_t^S, \qquad dv_t=\kappa(\theta-v_t)\,dt+\xi\sqrt{v_t}S_t\,dW_t^v.$$

v_t is variance, κ reversion speed, θ long-run variance, ξ its volatility coefficient and ρ Brownian correlation. We use the assignment's variance diffusion, including S_t. Inputs are $S_0 = 1$, $v_0 = 0.0064$ and μ = r = q = 0; baseline K = T = 1, θ = v_0, κ = 2, ξ = 0.04 and ρ = 0.5.

Both Euler updates use the old state; negative variance proposals are set to zero before the next square root. Scenarios satisfy the specified $2\kappa\theta > \xi^2$. Continuation uses ϕ = (1, x, z, x^2, xz, z^2), x = S/K and z = v/v_0, capturing variance and its interaction with spot. Figure 1.2 varies strike, maturity and θ with initial variance fixed.

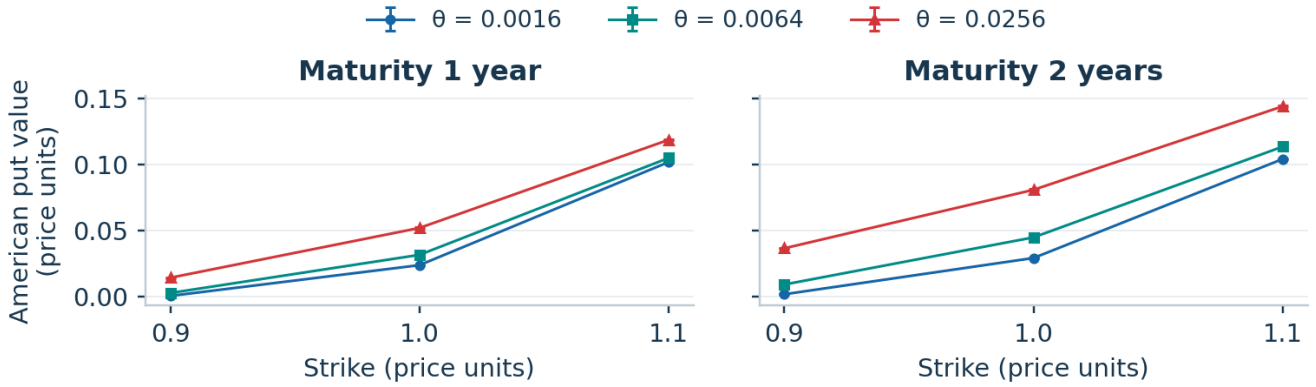


Figure 1.2. Daily-step put prices for 18 contracts: three strikes, two maturities and three long-run variances. Prices use the underlying's units ($S_0 = 1$); bars show ±2 evaluation SEs.

Prices increase with the tested strike, maturity and long-run variance, preserving the expected economic relationships. Repeating all 18 contracts at coarser steps gives Table 1.4: absolute changes from corresponding daily prices. These measure timestep sensitivity; no exact reference is supplied.

Table 1.4. Mean absolute price change from daily Euler steps (price units).

Long-run variance θ	2-day change	4-day change	7-day change
0.0016	0.000149	0.000144	0.000054
0.0064	0.000131	0.000096	0.000235
0.0256	0.000212	0.000227	0.000240

For example, 0.000240 averages the absolute daily-to-weekly changes across six strike/maturity contracts at θ = 0.0256. It is an average, not the largest individual difference.

The largest individual change is 0.000497 at K = T = 1, θ = 0.0256: 0.052394 daily versus 0.051897 weekly. Of 18 contracts, 17 changes lie within two combined evaluation SEs. Variance projection occurs once at daily steps and 30 times at weekly steps; no negative spots occur. Section 3.1 explains boundary treatment.

1.3 - Four stochastic-volatility sensitivities

How the variance parameters change the American put value

The long-run variance level has the strongest observed effect. Reversion speed and variance volatility have smaller effects around this baseline.

Vary one parameter at a time for the one-year put: $S_0 = K = 1$, $v_0 = \theta = 0.0064$, $\kappa = 2$, $\xi = 0.04$, $\rho = 0.5$ and zero rates. Baseline value: 0.03165 price units. Shared random draws improve comparison precision. Figure 1.3 plots changes from this baseline; Table 1.5 interprets them.

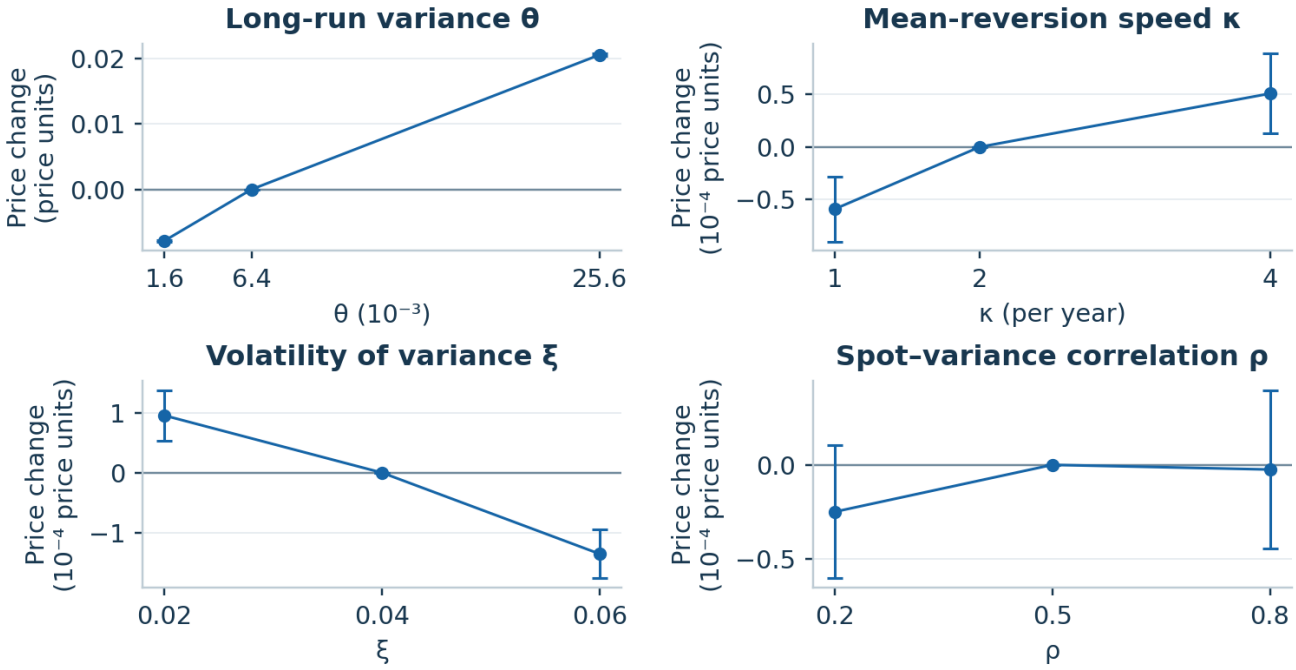


Figure 1.3. Price change versus each parameter; bars show ±2 paired SEs. The θ panel uses price units, the others 10^{-4} price units. One price unit equals initial spot.

Table 1.5. Economic interpretation of the sensitivity results in Figure 1.3.

Parameter	Economic interpretation and observed result
Long-run variance θ	θ = 0.0016 → 0.0256 raises value from 0.02382 to 0.05216 price units. Greater expected future fluctuations make protection against low spot prices more valuable.
Reversion speed κ	κ = 1 → 4 raises value from 0.03159 to 0.03170 price units. Faster reversion shortens variance deviations; the effect is small because initial variance equals its long-run level.
Volatility of variance ξ	ξ = 0.02 → 0.06 lowers value from 0.03174 to 0.03151 price units. A higher ξ broadens variance fluctuations while θ fixes their long-run mean. Here this slightly reduces the put value; the change exceeds paired sampling uncertainty.
Correlation ρ	Positive ρ links spot and variance shocks. Values at ρ = 0.2/0.5/0.8 are 0.03162/0.03165/0.03165 price units. Both baseline changes are below two paired SEs: the test cannot distinguish this small effect from sampling noise.

Long-run variance is the main value driver here. Further independent fits and evaluations could establish whether the small correlation effect is reproducible. The plotted bars measure evaluation uncertainty with fitted rules fixed.

2.1 - European control variate

The European control reduces price-estimator variance by roughly three quarters at the same path count.

We price one-month American puts under GBM: S_0=K=40, r=q=0, scaled quadratic basis, daily days/365 steps and a shorter final interval. For each volatility, 50,000 training paths estimate the exercise rule; 100,000 independent evaluation paths follow it to estimate the price. More evaluation paths improve precision without further regressions.

American payment X=$(K$-$S_\tau^{\text{Euler}})^+$ follows the fitted exercise rule; European payment Y=$(K$-$S_T^{\text{exact}})^+$ uses the same Brownian increments:

$$S_T^{\text{exact}}=S_0\exp\left(-\frac{1}{2}\sigma^2T+\sigma\sum_j \Delta W_j\right), \qquad m_Y=\mathbb{E}[Y]=p_{\text{BS},0}.$$

$$p_{\text{BS},0}=K\Phi(-d_2)-S_0\Phi(-d_1), \quad d_1=\frac{\ln(S_0/K)+\sigma^2T/2}{\sigma\sqrt{T}}, \quad d_2=d_1-\sigma\sqrt{T}.$$

BS denotes Black–Scholes; Φ is the standard normal cumulative distribution. The exact price $p_{\text{BS},0}$ supplies the control mean. Estimate β on training payments and fix it for evaluation. The control-variate (CV) adjustment is:

$$Z_i=X_i-\hat{\beta}(Y_i-m_Y), \qquad \hat{P}_{\text{CV}}=\overline{Z}, \qquad \beta^*=\frac{\text{Cov}(X,Y)}{\text{Var}(Y)}.$$

Z_i is path i's adjusted payment. For any fixed β, $E[Z]$=$E[X]$-$\beta(E[Y]$-$m_Y)$=$E[X]$: the correction has zero mean. With ρ_{XY} denoting payment correlation, optimal variance is $\text{Var}(X)(1-\rho_{XY}^2)$. The estimated variance of the average is s_Z^2/n: s_Z is the sample standard deviation of adjusted payments and n=100,000. Its square root is the standard error (SE).

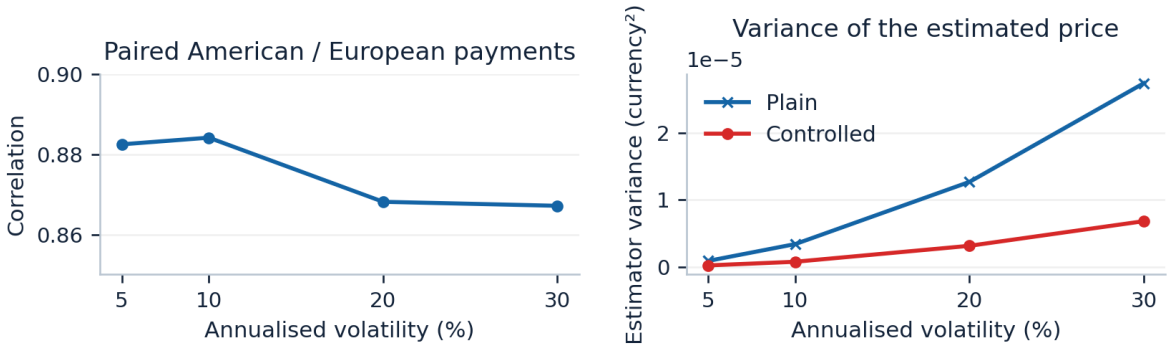


Figure 2.1. One-month American put: payment correlation and price-estimator variance at 5%, 10%, 20% and 30% volatility. Variance: currency units squared.

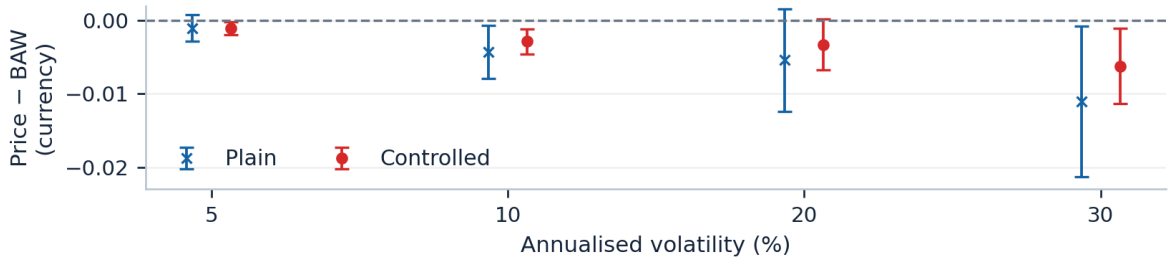


Figure 2.2. One-month American put errors versus volatility (currency units), against the Barone-Adesi and Whaley reference derived below. Bars: ±1.96 SE, fitted rule and coefficient fixed.

Payment correlation of 0.87–0.88 roughly halves price standard error, improving valuation precision.

2.2 - Barone-Adesi and Whaley comparison

At zero rates and yields, the American put equals the European put: we have an exact benchmark for the implementation.

Barone-Adesi and Whaley (1987), abbreviated BAW, write $P=p+\varepsilon$. Here ε is the American put's early-exercise premium above European value p. Their continuation equation, with time to maturity T, is:

$$\varepsilon_T=\frac{1}{2}\sigma^2S^2\varepsilon_{SS}+(r-q)S\varepsilon_S-r\varepsilon.$$

Following the article, set $H=1-e^{-rT}$ and $\varepsilon=Hf(S,H)$. Then $H_T=r(1-H)$ and $\varepsilon_T=r(1-H)(f+Hf_H)$. Substitute and divide by $H\sigma^2/2$:

$$S^2f_{SS}+NSf_S-\frac{M}{H}f-M(1-H)f_H=0,\quad M=\frac{2r}{\sigma^2},\quad N=\frac{2(r-q)}{\sigma^2}.$$

BAW drops the last term, which disappears at short and long maturities; the paper also checks accuracy numerically. The resulting ordinary differential equation admits $f=aS^\lambda$, with $\lambda^2+(N-1)\lambda-M/H=0$. The negative root ensures the put premium vanishes as spot rises:

$$\lambda=\frac{-(N-1)-\sqrt{(N-1)^2+4M/H}}{2}<0.$$

The continuation value is $p(S,T)+A_B(S/B)^\lambda$. At exercise threshold B, matching the value and slope to the immediate payoff gives:

$$K-B=p(B,T)+A_B,\qquad -1=p_S(B,T)+\lambda A_B/B.$$

The European put delta, $p_S=-e^{-qT}\Phi(-d_1)$, determines the premium coefficient:

$$A_B=-\frac{B}{\lambda}\left[1-e^{-qT}\Phi\left(-d_1(B)\right)\right],\quad d_1(B)=\frac{\log(B/K)+(r-q+\sigma^2/2)T}{\sigma\sqrt{T}}.$$

We solve value matching by bisection in B/K. Our zero-rate application takes the article's formula to its limit: $q=0$, $r\to0$, $H\sim rT$, so:

$$M/H\to\frac{2}{\sigma^2T},\qquad \lambda_0=\frac{1-\sqrt{1+8/(\sigma^2T)}}{2}<0.$$

Setting $H=0$ directly would divide by zero. Zero interest removes the benefit of receiving the strike early: martingale spot and Jensen give $E[(K-S_T)^+|\mathcal{F}_t]\geq(K-S_t)^+$. Put–call parity gives $p(B,T)=K-B+c(B,T)>K-B$ for positive spot, time and volatility. No nonnegative premium can meet the payoff at $B>0$, so $B=0$. The term DS^{λ_0} diverges at zero unless $D=0$, whereas the put cannot exceed its strike. Hence $P_{\text{BAW},0}=p_{\text{BS},0}$, also the exact American value.

Volatility	Exact value	Plain error ± 1.96 SE	Controlled error ± 1.96 SE
5%	0.23033	-0.00104 ± 0.00182	-0.00107 ± 0.00086
10%	0.46064	-0.00427 ± 0.00361	-0.00287 ± 0.00169
20%	0.92119	-0.00540 ± 0.00699	-0.00330 ± 0.00347
30%	1.38154	-0.01102 ± 0.01028	-0.00623 ± 0.00512

Table 2.1. One-month put values and pricing errors across volatility (currency units). Error is estimate minus exact value; negative means undervaluation. Intervals show evaluation uncertainty, with the fitted rule and coefficient fixed.

Controlled prices are 0.36–0.62% low; three intervals exclude zero. This identifies a small residual Euler/exercise-decision error for Exercise 3. Our general BAW implementation also reproduces all five published Table IV prices to two decimals (maximum difference 0.0033): $K=100$, $T=0.25$, $\sigma=20\%$, $r=8\%$, $q=0$, spots 80–120.

2.3 - Gamma profile across moneyness

Gamma becomes markedly irregular in the 20% and 30% volatility cases. We measure this instability and investigate a larger spot bump as a practical improvement.

Gamma, $\partial^2 P/\partial S^2$, measures how hedge delta changes with spot. Keeping the one-month, zero-rate settings, we vary S/K from 0.90 to 1.10 at 21 equally spaced points around the near-strike curvature peak. The central-difference bump is h=0.2 currency units (0.5% of strike), with $O(h^2)$ error for smooth prices.

We fit separate exercise rules at starting spots S-h, S and S+h. Training draws are shared across spots, as are the independent evaluation draws, reducing noise in their differences. Apply each fitted rule unchanged; superscripts -, 0 and + identify its payment on path i:

$$G_{X,i}=\frac{X_i^{+}-2X_i^{0}+X_i^{-}}{h^2}, \qquad G_{Y,i}=\frac{Y_i^{+}-2Y_i^{0}+Y_i^{-}}{h^2}.$$

$G_{X,i}$ and $G_{Y,i}$ are the American and European gamma observations. Their European control mean is the corresponding finite difference of exact prices:

$$m_G=\frac{p_{\mathrm{BS},0}(S+h)-2p_{\mathrm{BS},0}(S)+p_{\mathrm{BS},0}(S-h)}{h^2}.$$

$$\hat{\beta}_\Gamma=\frac{\widehat{\mathrm{Cov}}_{\mathrm{train}}(G_X,G_Y)}{\widehat{\mathrm{Var}}_{\mathrm{train}}(G_Y)}, \quad G_{Z,i}=G_{X,i}-\hat{\beta}_\Gamma\left(G_{Y,i}-m_G\right), \quad \hat{\Gamma}_{\mathrm{CV}}=\overline{G}_Z.$$

"Train" means covariance and variance use training gamma observations. The adjusted evaluation observation is $G_{Z,i}$; its average gives CV gamma with β_Γ fixed. The analytical Black–Scholes reference is $\Gamma_{\mathrm{BS}}=\phi(d_1)/\left(S\sigma\sqrt{T}\right)$, where ϕ is the standard normal density and $d_1=[\log(S/K)+\sigma^2T/2]/\left(\sigma\sqrt{T}\right)$.

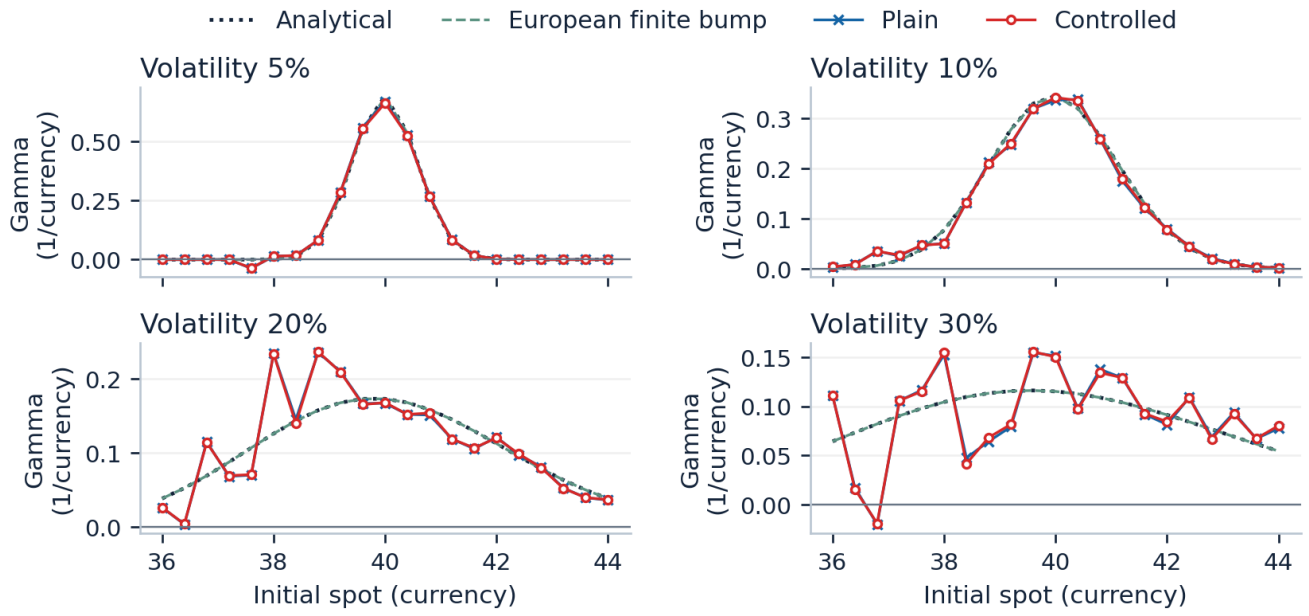


Figure 2.3. One-month American put gamma versus spot at 5%, 10%, 20% and 30% volatility (inverse currency units; separate scales). Analytical and European finite-bump references nearly coincide.

The lower-volatility profiles capture the near-strike peak more closely; the 20% and 30% profiles depart markedly from it. Negative estimates also occur at 5% and 30%. The control hardly changes the curves, directing our attention to bump size.

Gamma isolates curvature after the linear price response cancels. Division by the squared spot bump h^2 gives curvature per unit spot squared, but magnifies noise when bumps change fitted exercise dates. We next examine estimator variance and bump sensitivity.

2.3 - Gamma-estimator variance and stability

A larger bump improves the tested 20% at-the-money gamma estimate; the control alone has little effect.

Figure 2.4 compares gamma-estimator variance against spot at the four volatilities: s_G^2/n is the pathwise gamma variance divided by 100,000 paths. “Conditional” means fitted exercise rules and coefficient remain fixed.

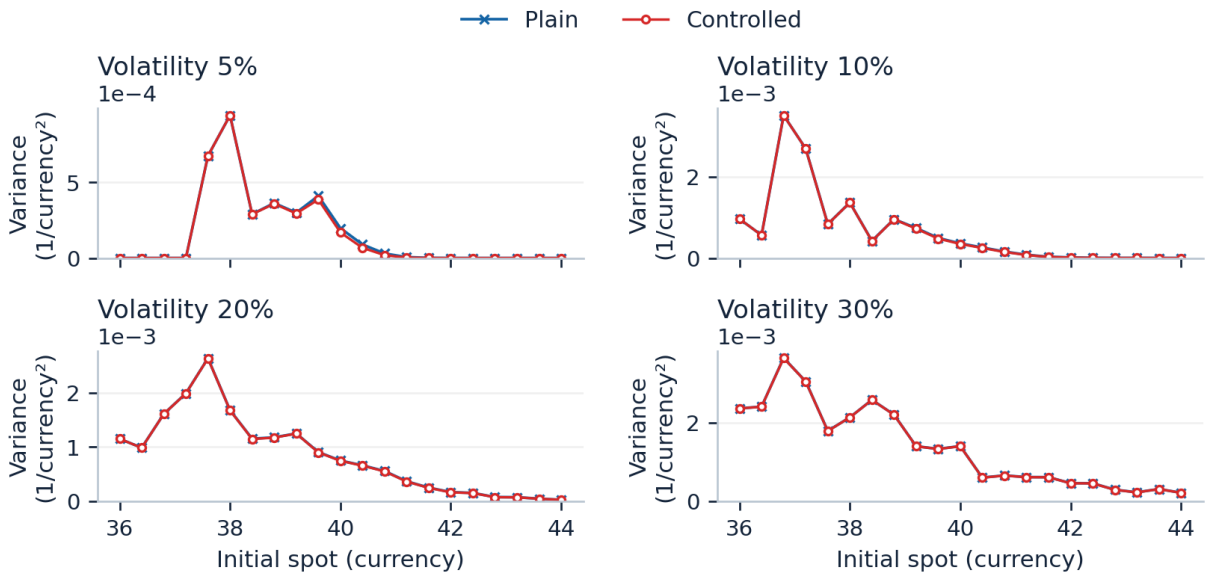


Figure 2.4. Plain and controlled gamma-estimator variance versus spot (inverse currency units squared; separate scales). Average variance reductions are only 0.18–3.01%.

a) Improving stability with the spot bump

Our existing results vary spot bump h=0.1,0.2,0.4 currency units at 20% volatility and S/K=0.98,1.00,1.02. Table 2.2 shows S=K=40, one month, zero rates. Bumps share 50,000 training and 100,000 evaluation draws; rules are refitted.

Bump h	Controlled gamma	Conditional variance
0.1	0.28920	5.57e-03
0.2	0.23123	7.00e-04
0.4	0.19224	8.92e-05

Table 2.2. Spot-bump sensitivity, 20% volatility, at the money. Units: currency for bump, inverse currency for gamma, inverse currency squared for variance. Analytical gamma: 0.172675.

Increasing h from 0.2 to 0.4 cuts conditional variance 7.8 times and observed absolute error by two thirds. Giles, Monte Carlo Evaluation of Sensitivities in Computational Finance (2007), Section 2.3, explains: small bumps amplify noise; larger bumps increase approximation error. The exact European finite-bump gamma at h=0.4 is 0.172250, only 0.25% below analytical gamma.

b) Variation across complete runs

Ten independent fits and evaluations at S=K=40, 20% volatility and h=0.2 give across-run variances 7.876×10^{-4} plain and 8.413×10^{-4} controlled. These include training variation for one complete estimate, not an average of ten.

The evidence supports h=0.4 for the 20% at-the-money case. Next, check the full spot profile and 30% volatility. Under zero-rate GBM, analytical European gamma remains the exact risk reference.

3.1 - Bias, variance and the Milstein correction

Milstein improves path convergence; we then measure what that improvement delivers for prices and gamma.

To separate numerical bias from sampling variance, let V be the American put value under optimal exercise. After fitting the exercise rule, our zero-rate estimator averages payoffs on N independent evaluation paths:

$$\hat{V}_{N,h}=\frac{1}{N}\sum_{i=1}^{N}\left(K-S_{\tau_i}^{(i)}\right)^{+}, \qquad m=\mathbb{E}\left[\hat{V}_{N,h}\right].$$

Here h is the time step and τ_i is the exercise date selected on path i. The mean m averages complete repetitions, including refitting. Adding and subtracting m makes the centred cross term vanish:

$$\mathbb{E}\left[\left(\hat{V}_{N,h}-V\right)^2\right]=\mathrm{Var}\left(\hat{V}_{N,h}\right)+(m-V)^2.$$

The first term is sampling variance; the second is squared bias. More paths and the European control target variance. Smaller steps target time-discretisation bias; fitted decisions, exercise dates and boundary conventions also affect bias. A single benchmark gap contains sampling noise, so we assess path convergence and pricing separately.

Retain the next stochastic Taylor term

Milstein (1975, Sections 2 and 5) retains the next stochastic Taylor term. For GBM, the diffusion derivative is $(\sigma S)\partial_s(\sigma S) = \sigma^2 S$. With h in years:

$$S_{t+h}=S_t+\mu S_t h+\sigma S_t \Delta W+\frac{1}{2}\sigma^2 S_t\left((\Delta W)^2-h\right).$$

The added term follows from the Itô identity $\int(W_u - W_t)dW_u = ((\Delta W)^2 - h)/2$. It captures the diffusion's movement within a step, which Euler omits. For smooth GBM coefficients, this raises the strong convergence order from one-half to one.

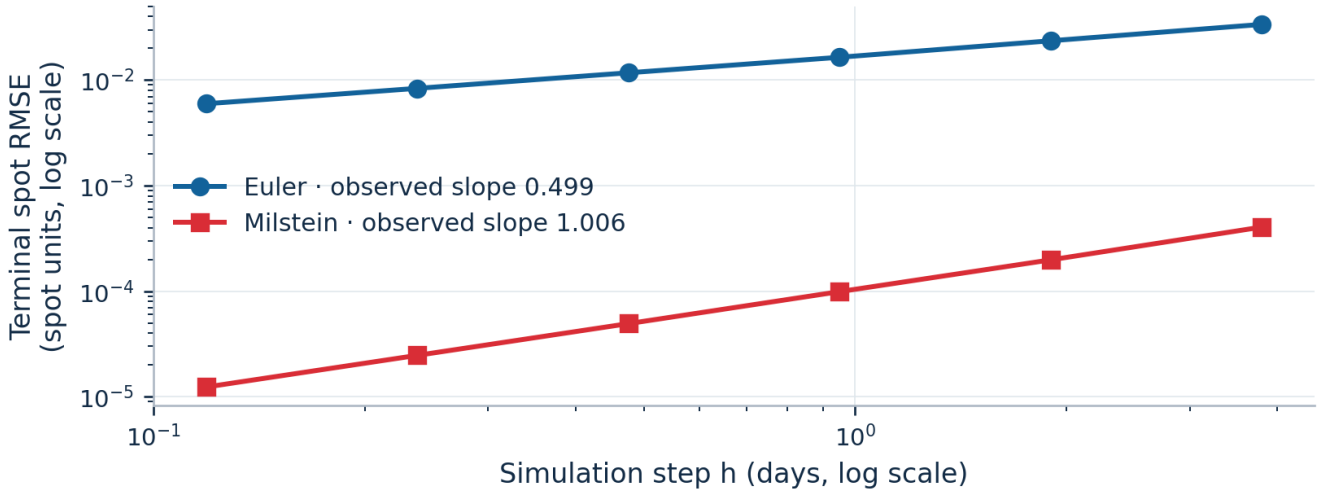


Figure 3.1. GBM terminal-spot root mean squared error (RMSE) against the exact transition on 10,000 shared Brownian paths: S_0 = 40, T = one month, σ = 20%. Coarser increments sum the finest increments; the six grids contain 8–256 steps. This isolates simulation accuracy before pricing.

The observed log–log slopes across the four volatilities are 0.499 for Euler and 1.005–1.006 for Milstein. Thus halving the step approximately halves Milstein's path error, compared with a factor of $1/\sqrt{2}$ for Euler. The implementation reproduces the expected convergence improvement.

3.1 - Deriving both state updates

The same principle gives corrections for both spot and variance, including their interaction.

Milstein's multidimensional construction (1975, Section 5) applies to both equations below. The variance diffusion contains S, so spot movements also affect the variance correction:

$$dS=\mu S\,dt+S\sqrt{v}\,dW^{S},\qquad dv=\kappa(\theta-v)\,dt+\xi S\sqrt{v}\,dW^{v}.$$

1- Expand the common diffusion factor

For v > 0, expand $S\sqrt{v}$ using the leading random changes $\Delta S \simeq S\sqrt{v}\,\Delta W^{s}$ and $\Delta v \simeq \xi S\sqrt{v}\,\Delta W^{v}$. This supplies Milstein's correction coefficients:

$$\Delta\left(S\sqrt{v}\right)\simeq\sqrt{v}\,\Delta S+\frac{S}{2\sqrt{v}}\,\Delta v\simeq Sv\,\Delta W^{S}+\frac{\xi S^{2}}{2}\,\Delta W^{v}.$$

Define $I_{ab}=\int_{t}^{t+h}(W_{u}^{a}-W_{t}^{a})\,dW_{u}^{b}$: the first subscript moves first. Integrating the expansion against each driver gives:

$$\text{spot correction}=SvI_{SS}+\frac{\xi S^{2}}{2}I_{vS},\quad \text{variance correction}=\xi SvI_{Sv}+\frac{\xi^{2}S^{2}}{2}I_{vv}.$$

With $I_{aa}=((\Delta W^{a})^{2}-h)/2$, the code advances both states from the same old pair (S, v):

$$S_{\text{next}}=S+\mu Sh+S\sqrt{v}\Delta W^{S}+\frac{1}{2}Sv((\Delta W^{S})^{2}-h)+\frac{1}{2}\xi S^{2}I_{vS}.$$

$$v_{\text{next}}^{\text{raw}}=v+\kappa(\theta-v)h+\xi S\sqrt{v}\Delta W^{v}+\frac{1}{4}\xi^{2}S^{2}((\Delta W^{v})^{2}-h)+\xi SvI_{Sv}.$$

2- Recover correlated integrals from independent drivers

Set $W^{S}=B_{1}$, $W^{v}=\rho B_{1}+\sqrt{1-\rho^{2}}B_{2}$, using independent increments of variance h. This gives covariance ρh. Write $J_{a\beta}$ for their integrals and $A = (J_{12} - J_{21})/2$:

$$I_{Sv}=\rho J_{11}+\sqrt{1-\rho^{2}}J_{12},\qquad I_{vS}=\rho J_{11}+\sqrt{1-\rho^{2}}J_{21}.$$

$$J_{11}=\frac{1}{2}((\Delta B_{1})^{2}-h),\quad J_{12}=\frac{1}{2}\Delta B_{1}\Delta B_{2}+A,\quad J_{21}=\frac{1}{2}\Delta B_{1}\Delta B_{2}-A.$$

$$c=\frac{1}{2}(\Delta W^{S}\Delta W^{v}-\rho h),\qquad I_{Sv}=c+\sqrt{1-\rho^{2}}A,\quad I_{vS}=c-\sqrt{1-\rho^{2}}A.$$

These identities are our application of Milstein's cross integrals to correlated drivers. Endpoints determine their sum; the stochastic area A supplies the remaining path information. We simulate A with Wiktorsson's established method, detailed next.

3- State boundary conventions and limiting checks

At v = 0, the code extends the simplified correction products by their finite interior limits, then sets $v_{\text{next}}=\max(v_{\text{next}}^{\text{raw}},0)$. This defines the next square root. Negative finite spots are counted without clipping; non-finite proposals fail.

Together, these steps implement both corrections, correlated integrals and boundary handling. Tests recover scalar GBM when ξ = 0 and $v_0 = \theta$, and the common-driver limit at ρ = 1; numerical differentiation independently checks the diffusion coefficients.

3.1 - Simulating and checking the cross integrals

Wiktorsson's method supplies the missing area while controlling its approximation cost.

Cross integrals contain path information beyond the Brownian endpoints. Wiktorsson (2001) supplies their established simulation method. We specialise it to the two independent endpoints $w_1 = \Delta B_1$ and $w_2 = \Delta B_2$:

1- Preserve endpoint dependence in the finite series

Specialising Wiktorsson's equation 2.2 to two drivers, draw independent standard normals X_{1k}, X_{2k}, Y_{1k}, Y_{2k} and set $\eta_i = \sqrt{2/h}\, w_i$. Retain n terms; a_n measures the omitted tail:

$$F_n = \sum_{k=1}^{n} \frac{X_{1k}(Y_{2k}+\eta_2) - X_{2k}(Y_{1k}+\eta_1)}{k}, \qquad a_n = \frac{\pi^2}{6} - \sum_{k=1}^{n} \frac{1}{k^2}.$$

Following the paper's page-481 algorithm, approximate that tail by a Gaussian G, independent of the retained terms conditional on w:

$$G|w \sim N\left(0,\ 2a_n\left[1+\frac{w_1^2+w_2^2}{h}\right]\right), \qquad \hat{A} = \frac{h}{2\pi}(F_n+G).$$

Reconstruct $J_{12} = w_1 w_2/2 + A$ and $J_{21} = w_1 w_2/2 - A$, then apply the correlated mapping above. This gives the code both required integrals with the correct endpoint sum. Retaining the series matters: the complete area is not Gaussian.

2- Choose the term count from the published error bound

$$\mathbb{E}\left|J_{12} - \hat{J}_{12}\right|^2 \le \frac{5h^2}{6\pi^2 n^2}, \qquad n(h) = \text{ceil}\left(\sqrt{\frac{5}{6\pi^2 h}}\right).$$

For two drivers, Theorem 4.1 and equation 4.9 give this bound averaged over endpoints. We choose a local mean-square budget h^3, with h in years: six terms per daily step and eight per half-day. Each path-step needs 4n + 1 extra normals, drawn separately so Euler and Milstein retain shared endpoints.

3- Check the conditional distribution and pricing sensitivity

For an independent distribution check, we use Wiktorsson's equation 1.5 with the known endpoint phase removed. This tests more than variance. Setting $x = \lambda h/2$ and $|w|^2 = w_1^2 + w_2^2$ gives:

$$\mathbb{E}\left[e^{i\lambda A}|w\right] = \frac{x}{\sinh x}\exp\left[-\frac{|w|^2}{2h}(x\coth x - 1)\right].$$

We compare simulated averages of $\exp(i\lambda A)$ with this expression at zero and nonzero endpoints, several frequencies, and six/twelve terms; we also verify the exact integral-sum identity. At $\lambda = 0$ the expression is one by continuity. Section 3.3 then checks the resulting option prices.

Together, these checks validate the coefficients and sampler used in the joint implementation. The local bound concerns integral error. Our projected boundary treatment and non-nested area tails mean the joint pricing refinements test numerical stability, rather than establish strong order one or polynomial-regression convergence.

3.2 - GBM pricing errors and the European control

The European control improves price precision under both simulators; daily Euler and Milstein prices are close.

We repeat Exercise 2 for an American put with $S_0 = K = 40$, one-month maturity, zero rates and yields, daily exercise, and volatilities 5%, 10%, 20% and 30%. The Barone-Adesi and Whaley (BAW) reference equals the exact European price here.

Each scheme fits its exercise rule and control coefficient on 50,000 training paths. Evaluation follows these rules on new paths, using the first 10,000, 25,000, 50,000, 75,000 and 100,000 paths of one sample. These estimates are dependent. Euler and Milstein share Brownian increments and the exact-GBM European control.

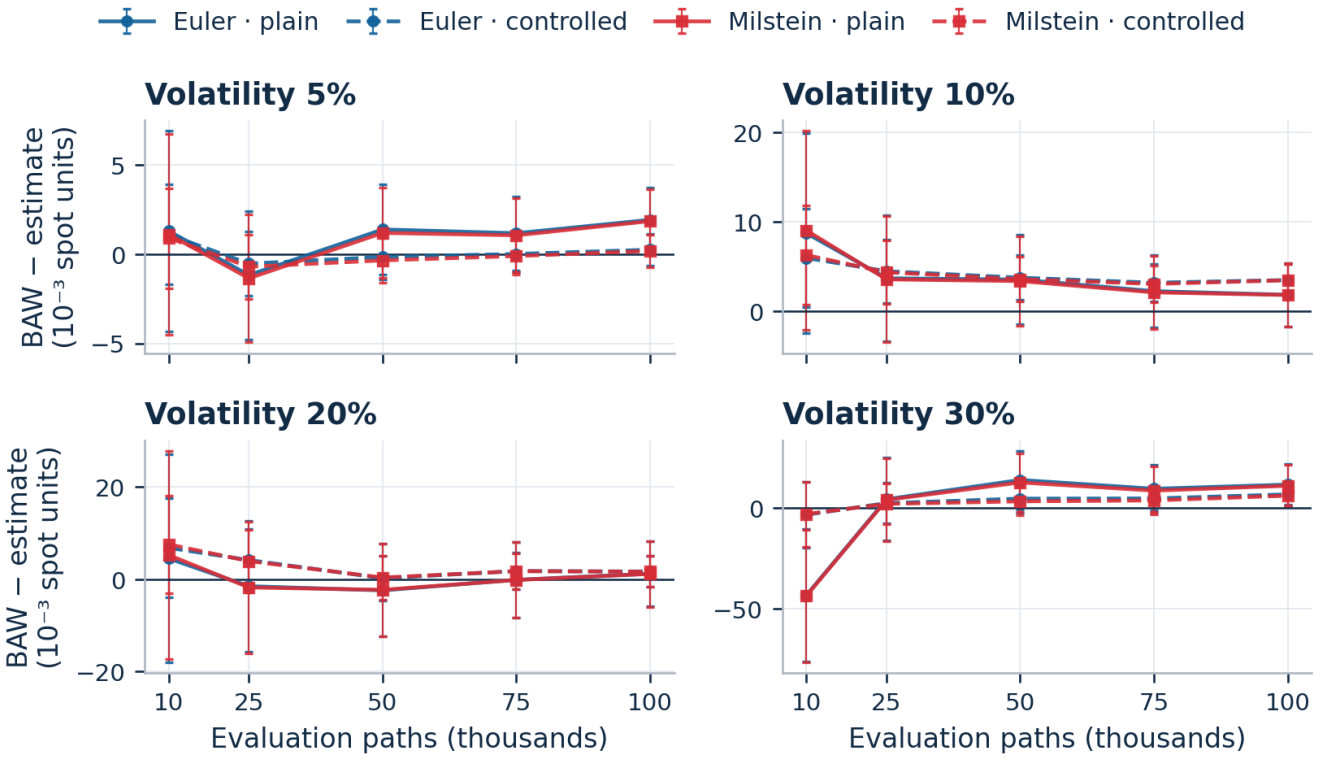


Figure 3.2. One-month American put pricing error against Barone-Adesi and Whaley over 10,000–100,000 evaluation paths, by volatility. Error = reference minus estimate; positive means undervaluation. Solid curves: plain estimates; dashed: controlled estimates. Bars: ±1.96 standard errors with fitted rules and coefficients held fixed.

Volatility	Reference price	Euler controlled error	Milstein controlled error
5%	0.230327	+0.000242	+0.000173
10%	0.460643	+0.003495	+0.003471
20%	0.921190	+0.001704	+0.001666
30%	1.381545	+0.006861	+0.006221

Table 3.1. Controlled one-month American put errors at 100,000 evaluation paths. The reference is the exact zero-rate price; all prices and errors are in spot units. Figure 3.2 shows the path-count comparison.

At 100,000 paths, the control reduces estimator variance by 3.9–4.5 times. Milstein's controlled errors are slightly smaller in all four cases, while paired plain-price differences remain within sampling uncertainty. The 10% and 30% controlled gaps exceed their evaluation-only intervals: the control improves precision, while fitted-decision and discretisation effects need separate assessment.

3.2 - Repeated gamma profiles and estimator variance

The repeated profiles point towards improving the gamma estimator, using the larger-bump evidence in section 2.3.

We repeat the American put gamma calculation with K = 40, one-month maturity, zero rates and a spot bump of 0.2 (0.5% of strike). Each of the three bumped valuations refits its rule on 50,000 paths using shared random draws. The rules and gamma control coefficient are then held fixed on 100,000 independent evaluation paths.

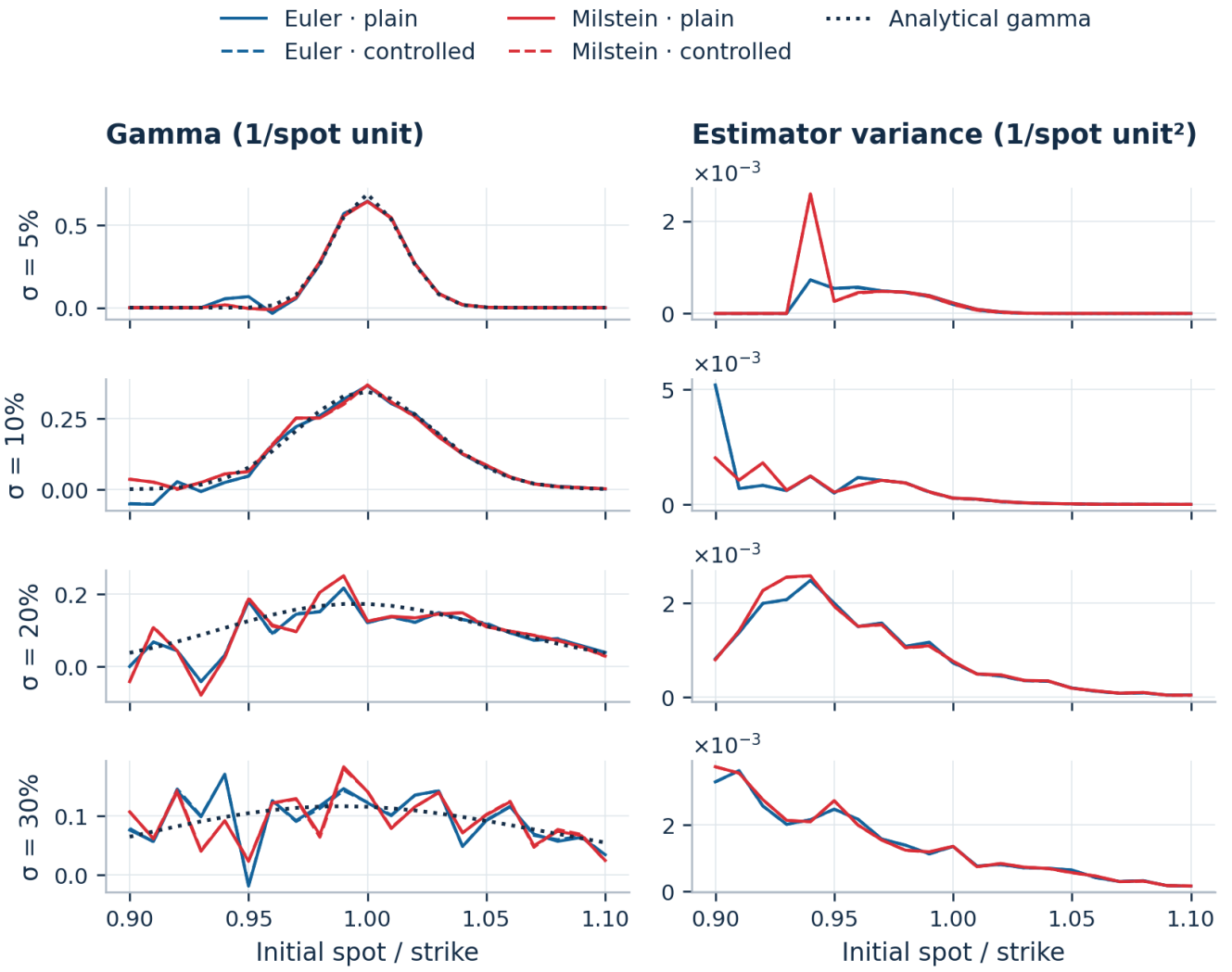


Figure 3.3. One-month American put gamma at four volatilities. Left: plain and controlled gamma against initial spot/strike, with exact zero-rate gamma dotted. Right: variance of the estimated gamma on 100,000 evaluation paths, holding the fitted rules and control coefficient fixed. Near-zero tail variances remain on linear axes.

At the money, the Milstein gamma control reduces conditional variance by approximately 15%, 4%, 0.8% and 0.1% as volatility rises from 5% to 30%. Its benefit is therefore concentrated at low volatility. At 20%, controlled gamma is 0.1258, compared with 0.1217 for Euler and 0.1727 analytically.

Milstein improves path accuracy, but the irregular gamma profiles persist at 20% and 30%. This identifies the sensitivity calculation as the next target. Section 2.3's existing Euler experiment found that doubling the bump reduced conditional variance 7.8-fold and moved gamma closer to the reference. Testing that refinement with Milstein is the natural next step; these profiles measure evaluation variance only.

3.3 - Joint-model prices, convergence and cost

The joint implementation is stable to the tested refinements; its extra cost brings no pricing advantage here.

We price the one-year American put at $S_0 = K = 1$, zero rates, $v_0 = \theta = 0.0064$, $\kappa = 2$, $\xi = 0.04$ and $\rho = 0.5$. We retain the six-term basis, use 50,000 training and 10,000–100,000 evaluation paths, daily exercise, and shared Brownian increments across schemes.

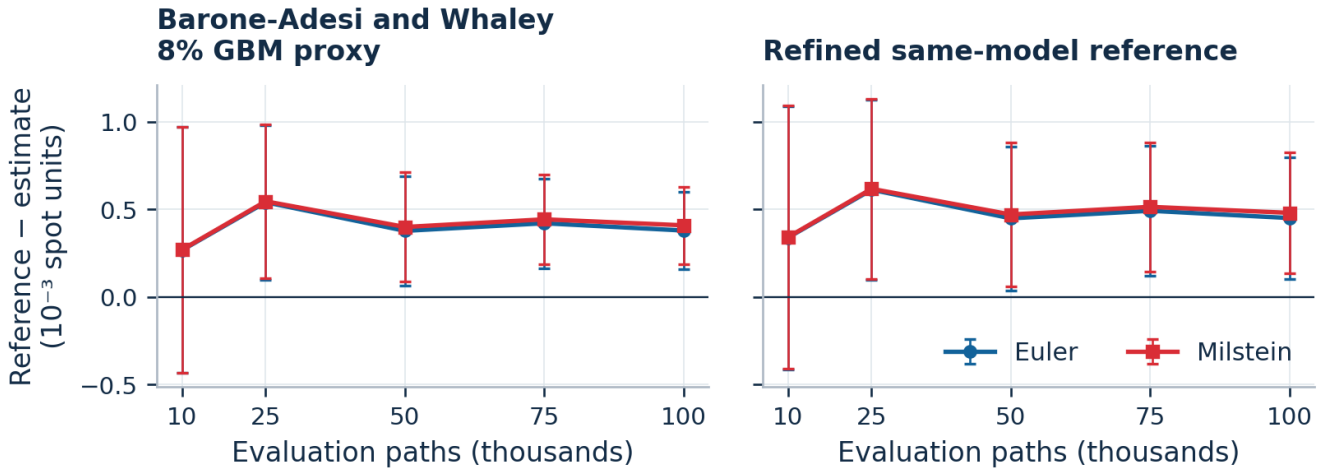


Figure 3.4. One-year joint-model put: reference minus LSMC price over 10,000–100,000 evaluation paths. Left: Barone-Adesi and Whaley at constant 8% GBM volatility. Right: refined terminal payments under the same joint model. Bars: ±1.96 standard errors, including reference uncertainty on the right.

Scheme	LSMC price	Joint-model reference − price	Runtime
Euler	0.031529	+0.000450	3.43 s
Milstein	0.031499	+0.000480	20.27 s

Table 3.2. One-year joint-model put, 100,000 evaluation paths. Gaps use the refined reference 0.0319782; values are in spot units. Runtime includes simulation, fitting and evaluation, excluding shared endpoint generation. Milstein takes 5.9 times longer.

The constant-8% comparison includes a model difference, so we also construct a joint-model numerical reference. For a well-defined nonnegative zero-drift spot process, conditional Jensen makes waiting optimal for this zero-rate put. Averaging 100,000 refined projected-Euler maturity payments gives 0.0319782 (SE 0.0001361). Refining ¼ to ⅛ day changes it by -4.92×10^{-7} (paired SE 6.39×10^{-7}).

In the separate 10,000-training/10,000-evaluation sensitivity check, doubling the daily Fourier terms from 6 to 12 changes the Milstein price by +0.0000287, against individual SEs near 0.000348. Half-day stepping changes it by −0.0000023. These small changes support the chosen daily sampling settings at the reported precision.

The main runs encounter no variance projections or negative spots. Both fitted rules exercise early on about 36% of paths, although waiting is optimal under the stated assumptions. Milstein lowers price by 0.0000298 (paired SE 0.0000101), moving farther from the reference. We have implemented and checked both state corrections; improving exercise decisions, and the bump choice for GBM gamma, is more promising here than extra stepping cost.

4.1 - Simulate the daily gas price

We implement the article's spot model on all daily decision dates and generate the assignment's 10,000 paths using its exact log-price transition.

In Boogert and de Jong's equation (22), the log price $X_t=\ln S_t$ follows an Ornstein–Uhlenbeck (OU) process. Applying Itô's formula gives the long-run log mean $m=\mu-\sigma^2/(2\kappa)$ and the exact simulation step:

$$\frac{dS_t}{S_t}=\kappa(\mu-\ln S_t)\,dt+\sigma\,dW_t, \qquad dX_t=\kappa(m-X_t)\,dt+\sigma\,dW_t.$$

$$X_{t+1}=m+(X_t-m)e^{-\kappa}+\sqrt{q(1)}Z_t, \quad q(h)=\frac{\sigma^2\left(1-e^{-2\kappa h}\right)}{2\kappa}, \quad S_{t+1}=e^{X_{t+1}}.$$

Here Z_t is standard normal. In daily units, $\kappa=0.05$, $\sigma=0.0945$ and $\Delta t=1$. The specified 7 July to 30 June period, using 2005–2006, contains 359 operating dates and 358 price transitions. Rates and bid–ask spreads are zero.

We set $S_0=20$ and $\mu=\ln 20+\sigma^2/(4\kappa)$, giving stationary mean spot EUR 20/MWh, near the midpoint of the article's EUR 14.88–25.44 forward-price range. The permitted constant mean provides an illustrative price level. Figure 4.1 shows three paths from our simulation to explain mean reversion and antithetic pairing.

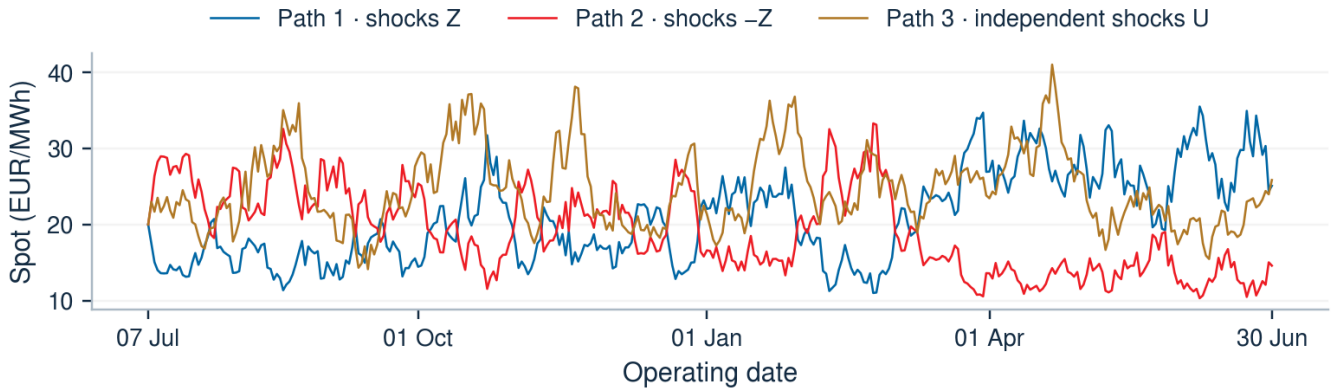


Figure 4.1. Daily simulated spot prices at 9.45% volatility. Paths 1 and 2 use opposite shocks; path 3 shows another independent shock sequence. Prices fluctuate around a stable level, while the paired paths move in opposite directions in log price.

Following the article, opposite shocks form 5,000 independent antithetic pairs. To check the simulator, we compare sampled log-price means and variances with their exact values: $E[X_t]=m+(\ln S_0-m)e^{-\kappa t}$ and $\mathrm{Var}(X_t)=q_t$.

Table 4.1. Simulated log-price moments versus the exact model moments.

Check at days 1, 90, 180 and 358	9.45% volatility	3.15% volatility
Largest absolute log-mean error (dimensionless)	4.44×10^{-16}	4.44×10^{-16}
Largest log-variance discrepancy (SE units)	1.17	3.08

The paired sample reproduces the log mean to rounding. The largest variance discrepancy is 3.08 standard errors in the low-volatility sample. The exact transition avoids time-discretisation error at daily decisions, so a Milstein correction is unnecessary here.

4.2 - Continuation values and feasible decisions

Storage adds an inventory state, repeated trading and several feasible volumes at each date. We handle this complexity by separating contract constraints, price simulation and backward valuation, while sharing regression calculations across inventories.

We compare the cash received or paid today with expected later cashflows. Inventory v_t is measured before trading; Δv is positive for injection and negative for withdrawal. With spot S_t in EUR/MWh and inventory in MWh, the zero-cost cashflow is:

$$v_{t+1}=v_t+\Delta v, \qquad h(S_t,\Delta v)=-S_t\Delta v.$$

Table 4.2. Contract and volume choices used in the implementation.

Contract input	Implemented value / source
Capacity; initial and final inventory	250,000; 100,000 and 100,000 MWh — assignment / Exhibit 1
Maximum daily injection / withdrawal	2,500 / 7,500 MWh — Exhibit 1
Inventory spacing and states	2,500 MWh; 101 levels — article's standard contract
Candidate daily trades (MWh)	−7,500; −5,000; −2,500; 0; +2,500

a) Enforce the contract before comparing actions

The article's equations (8) and (9) give the capacity and daily flow constraints. Our feasible set $D_t(v)$ also ensures that the final 100,000 MWh can still be returned. With n trading dates left, including today:

$$\max(0,100000-2500n)\le v_t\le\min(250000,100000+7500n).$$

Each trade must leave $v_t+\Delta v$ feasible with n-1 dates remaining. We index 7 July by t=0 and 30 June by $t=T$. Settlement follows that final trade. Equations (11) and (20) set continuation to zero at v_{T+1}=100,000; unreachable states are excluded from action selection and regression.

b) Compare immediate cashflow and continuation

Let U_t be the storage value before today's trade. Equation (10) values the remaining operating right after trading. Equation (12) adds the immediate cashflow and chooses the most valuable feasible volume:

$$C_t(s,v,\Delta v)=\mathbb{E}[U_{t+1}(S_{t+1},v+\Delta v)|S_t=s],$$
$$U_t(s,v)=\max_{\Delta v\in D_t(v)}\{-s\Delta v+C_t(s,v,\Delta v)\}.$$

Equation (16) simplifies continuation to $C_t(s,w)$, with resulting inventory $w=v+\Delta v$. Two actions reaching the same w share the same future opportunities, so the code reuses one fitted function. Their immediate cashflows still differ. Continuation therefore covers all later trading, rather than only tomorrow's payment.

The assignment supplies 100,000 MWh initially and requires its return; it specifies no initial cash-budget constraint. We value the operating right before unspecified storage rent. Daily flow limits are fixed as in Exhibit 1, and trading uses day-ahead spot prices.

4.2 - Fit, choose and carry actual cashflows

We implement the article's backward recursion: fit future value, choose a feasible trade, then carry the resulting realised cashflow to the preceding date.

c) Fit continuation across paths — equations (15) and (17)

At date t, fix feasible next inventory w. Later decisions are already estimated. On path b, $Y_{t+1}^{b}(w)$ is the cashflow earned from tomorrow onward. Ordinary least squares (OLS) fits these outcomes against today's spot across all M training paths:

$$\hat{C}_t(s,w)=\sum_{q=1}^{6}\hat{\beta}_{q,t,w}\,\phi_{q,t}(s), \qquad \Phi_{bq}=\phi_{q,t}\left(S_t^b\right).$$

$$\hat{\beta}_{t,w}=\text{argmin}_{\beta\in\mathbb{R}^6}\|Y_{t+1}(w)-\Phi\beta\|^2.$$

The article's b_6 basis consists of six cubic B-splines. OLS estimates their six weights, with no extra intercept. Knots follow each date's lognormal price distribution: interior quantiles 1/3 and 2/3; endpoints 0.0001 and 0.9999 repeated four times. Endpoint tangents extend the functions linearly beyond this range.

One paths × 6 matrix is factorised per date by singular value decomposition (SVD), solving all inventory regressions together. Action comparisons are vectorised across paths and inventories. Unlike an American put, storage can benefit from an immediate cash outflow through injection, and trading continues afterwards. Every price path therefore enters OLS: there is no in-the-money filter.

d) Choose the feasible trade — equation (18)

$$\widehat{\Delta v}_t^{\,b}=\text{argmax}_{\Delta v\in D_t(v)}\{-S_t^b\Delta v+\hat{C}_t\left(S_t^b,v+\Delta v\right)\}.$$

We score each allowed withdrawal, hold and injection at today's spot, then select the largest total. Inventory feasibility is checked first, so a profitable-looking trade cannot breach the contract.

e) Carry realised cashflows — equations (19) to (21)

$$Y_t^b(v)=-S_t^b\widehat{\Delta v}_t^{\,b}+Y_{t+1}^b\left(v+\widehat{\Delta v}_t^{\,b}\right), \qquad \hat{V}_{\text{fit}}=\frac{1}{M}\sum_{b=1}^{M}Y_0^b\,(100000).$$

Equation (18) chooses the highest total score. The backward update carries today's payment plus the same path's realised later cashflow. Separate current- and next-date arrays preserve those regression targets. Averaging inception cashflows gives V_{fit}. At the common initial spot, continuation is a sample mean; after the final trade, it is zero.

We then evaluate the rule on independently generated paths, holding its coefficients fixed. Each new path chooses trades from its own current spot and inventory. Averaging those realised operating cashflows measures how the learned decisions perform on new scenarios.

All 357 non-degenerate fits retain rank six, with maximum condition number 70.8. At most 8 of 10,000 high-volatility training paths use a linear tail on any date. These checks support stable regression; section 4.3 independently assesses decision quality.

4.2 - One complete storage decision

One saved training path connects the fitted decision, actual trading cashflows and the target used in the preceding regression.

On 29 June 2006, saved training path 1 has spot EUR 15.8802/MWh in the 9.45% case. At inventory 100,000 MWh, Table 4.3 compares all five trades using continuation fitted to 10,000 final-day cashflows. On the last date, purchases or sales must restore 100,000 MWh.

Table 4.3. Feasible actions on 29 June at 100,000 MWh: saved training path 1.

Trade today MWh	Next inventory MWh	Cashflow now EUR	Fitted future EUR	Total score EUR
−7,500	92,500	—	—	Rejected
−5,000	95,000	—	—	Rejected
−2,500	97,500	39,700.52	−40,261.68	−561.16
0	100,000	0.00	0.00	0.00
+2,500 (chosen)	102,500	−39,700.52	40,261.68	561.16

Injection wins: expected sale proceeds exceed today's purchase cost by EUR 561.16. Larger withdrawals are infeasible because tomorrow permits only 2,500 MWh of injection. Calculations use unrounded prices.

What is carried back to the preceding regression?

On this same path, the final-day spot is EUR 16.9450/MWh. Selling 2,500 MWh earns EUR 42,362.47 against today's EUR 39,700.52 purchase. This future outcome supplies a training target; it is unavailable to today's decision.

$$Y_{T\text{-}1}^{1}(100{,}000)\text{=-}39{,}700.524\text{+}42{,}362.466\text{=}2{,}661.942 \quad \text{EUR.}$$

Withdrawal restores 100,000 MWh. The realised EUR 2,661.94 becomes path 1's 28 June target for continuation at inventory 100,000 MWh. Thus the fitted score chooses the trade; the realised result supplies the preceding regression, including all later cashflows.

Check the decision against an exact two-day value

On 30 June, the only feasible trade restores the final inventory, giving $Y_T(w)\text{=}(w\text{-}100{,}000)S_T$. The known conditional mean of tomorrow's lognormal price therefore supplies an exact two-day continuation against which to check our regression:

$$C_{T\text{-}1}(s,w)\text{=}(w\text{-}100000)\exp\left[m\text{+}(\ln s\text{-}m)e^{-\kappa}\text{+}\frac{q(1)}{2}\right].$$

Table 4.4. Exact near-expiry and physical-constraint checks.

Implementation check	Measured result
Final-day inventories	97,500 to 107,500 MWh
Two-day spot quantiles	1%, 5%, 25%, 50%, 75%, 95%, 99%
Largest continuation discrepancy	EUR 118.73
Largest selected-action value loss versus exact optimum	EUR 0.00
Terminal inventory on every evaluated main-case path	100,000 MWh

All seven tested spots in each volatility case select the exact best trade: EUR 0 loss means no value lost through the decision, rather than zero trading cashflow. Every evaluated path respects capacity, flow limits and final inventory. These checks validate contract handling and the near-expiry recursion.

4.3 - Storage value, cashflow spread and convergence

The implemented strategy values storage at EUR 4.49m and EUR 1.45m. Greater price variability increases the value of adapting injection and withdrawal to observed prices.

We train on 10,000 paths and value the unchanged decision rule on 10,000 new paths. The 9.45% and 3.15% daily-volatility comparison follows Exhibit 3; both cases retain initial and stationary mean spot EUR 20/MWh.

Table 4.5. Full-year storage values and sampling uncertainty.

Daily volatility	Training value EUR	New-path value EUR	95% half-width EUR
9.45%	4,508,663	4,490,526	24,565
3.15%	1,454,190	1,445,649	7,681

Intervals are the new-path value ± the stated half-width, using 5,000 independent antithetic pair means and unchanged fitted coefficients.

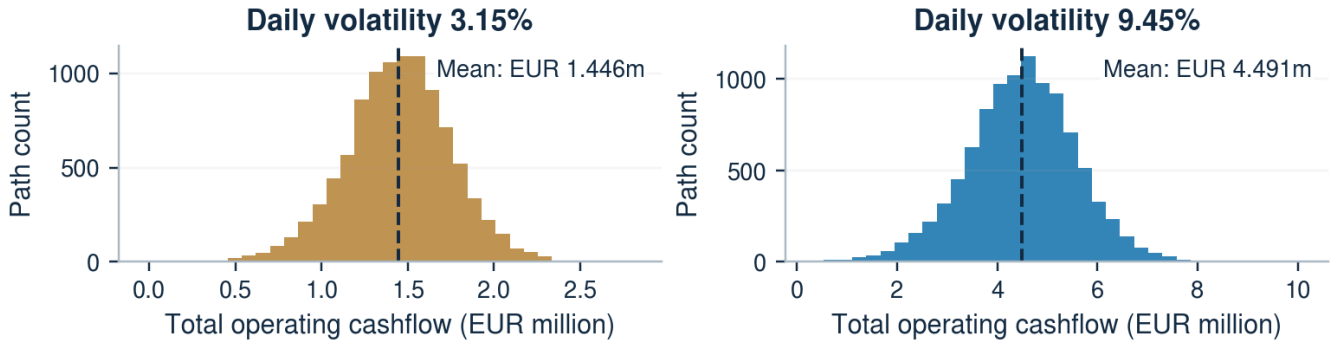


Figure 4.2. Distribution of total operating cashflow on 10,000 new paths; dashed lines mark means. Higher volatility produces both greater average value and a wider range of outcomes, answering the Exhibit 3 comparison. This cashflow spread differs from uncertainty in the estimated mean.

We measure the benefit of daily adaptation against the best schedule fixed at inception, calculated from the expected spot curve. Its expected cashflow is EUR 23,408 at high volatility and EUR 2,588 at low volatility. Responding to observed prices therefore generates most operating value in this constant-mean model.

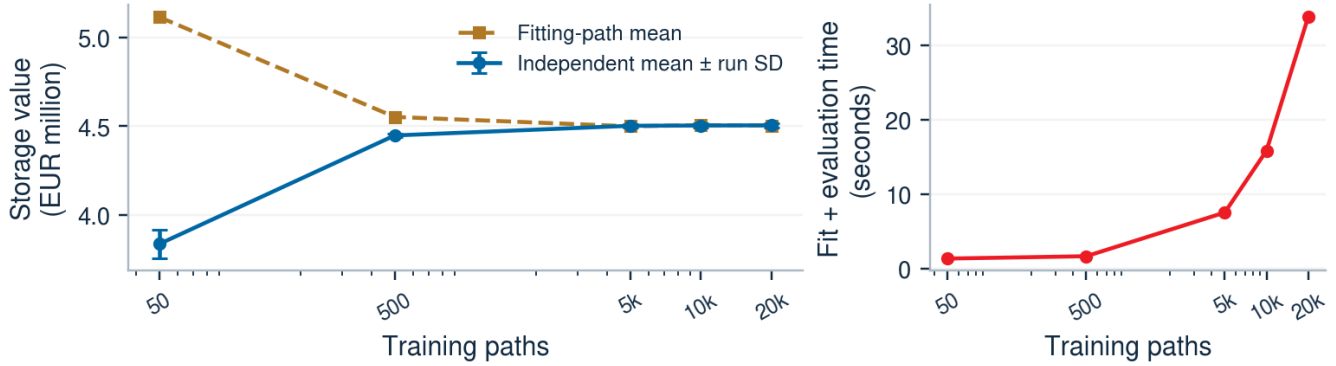


Figure 4.3. High-volatility training-path convergence: ten fits per count, sharing 10,000 new evaluation paths within each repetition. Left: mean value and one-run standard deviation. Right: mean fitting plus evaluation time with four workers.

Doubling training paths from 10,000 to 20,000 increases mean new-path value from EUR 4,503,747 to EUR 4,504,619: EUR 873 (0.019%; paired SE EUR 94), at about twice the runtime. We retain 10,000 as a good value–cost balance and next assess decision accuracy independently.

4.3 - Independent benchmark and finer trading volumes

Our independent reference places fitted decisions within about 0.04% of the optimum in both volatility cases.

No full-year benchmark is supplied for our constant-mean case. Backward Gauss–Hermite integration uses the same model and contract, replacing sampling with normal nodes z_j and weights p_j summing to one:

$$s_j(s)=\exp\{m+(\ln s-m)e^{-\kappa}+\sqrt{q(1)}z_j\},\quad C_t(s,w)\approx\sum_j p_j\,\tilde{U}_{t+1}\big(s_j(s),w\big).$$

The tilde means linear spot interpolation and endpoint extrapolation. The log-uniform grid covers initial and stationary means, including S_0. We refine spacing, quadrature and domain:

Table 4.6. Refinement of the independent optimised reference.

Price points[1]	Quadrature nodes	Tail width (log-price SD)	9.45% value EUR	3.15% value EUR
801	128	6	4,508,299	1,451,554
1,601	128	6	4,507,622	1,451,334
3,201	128	6	4,507,454	1,451,279
3,201	64	6	4,507,403	1,451,269
3,201	256	6	4,507,483	1,451,287
4,243 / 4,259	256	8	4,507,483	1,451,287

[1] Final counts show high / low volatility; all feasible trades are optimised.

Last spacing, quadrature and domain refinements change high-volatility value by EUR 169, EUR 29 and EUR 0.03, supporting stability.

Table 4.7. Fitted decisions versus the independently optimised reference.

Quantity (EUR)	9.45% volatility	3.15% volatility
Optimised numerical reference	4,507,483	1,451,287
Fitted decisions valued by integration	4,505,786	1,450,732
Value difference from optimised reference	1,697	555
Monte Carlo value of those decisions	4,490,526	1,445,649

Integrating saved decisions without refitting gives gaps of EUR 1,697 and EUR 555, moving at most EUR 7.18 in the last refinements. This independent benchmark establishes decision quality under this model and grid. Monte Carlo values agree within 1.22 and 1.30 sampling SE.

Finer-grid check: 1,250 versus 2,500 MWh

Halving inventory spacing to 1,250 MWh gives 201 states and nine trades, including partial injection. Both grids share training and evaluation paths, with separate medium-price, fine-quadrature and wide-domain benchmarks.

Table 4.8. Finer inventory spacing versus 2,500 MWh.

Finer grid minus baseline	9.45% volatility	3.15% volatility
Optimised change at all three settings (EUR)	0.00	0.00
Integrated fitted-rule change (EUR)	−16.22	−2.34
Paired Monte Carlo change ± SE (EUR)	−129.93 ± 227.36	144.36 ± 68.49
Fitting time: finer / original	3.55	3.60

We retain 2,500 MWh spacing: finer choices add no optimised value but require about 3.6 times the fitting time. The original grid is sufficient for these linear cashflows and fixed limits.

4.3 - Optional extension: Sobol and a mean-reverting bridge

We combined scrambled Sobol sampling with a Brownian bridge adapted to the mean-reverting price model. In the high-volatility case, this reduced valuation root-mean-square error by 74% at 16,384 paths, compared with antithetic Monte Carlo at similar cost.

a) Why we targeted sampling precision

Independent integration placed our fitted strategy EUR 1,697 below the optimised reference, while its Monte Carlo valuation had a standard error of EUR 12,533. We therefore tested how to value the same strategy more precisely. The daily price transition is already exact.

b) What scrambled Sobol contributes

Scrambled Sobol is a randomised quasi-Monte Carlo method. Its points cover the space of simulation inputs more evenly than independent random draws. Scrambling preserves that structure while allowing independent repetitions from which to estimate uncertainty [7].

c) What the bridge contributes

A Brownian bridge describes Brownian motion conditional on its endpoints. To construct a path, we first draw its final value, then fill intermediate dates conditional on the surrounding values already drawn following the Brownian-bridge construction illustrated in Mike Giles's Oxford lecture slides on quasi-Monte Carlo [8, slides 29 and 30]. This preserves the path distribution.

The methods work together: Sobol spreads the input points, while bridge ordering assigns its early coordinates to widely separated dates before adding daily detail. This can improve sampling for a payoff that depends on the whole path; our storage comparison tests the benefit.

d) How we implement the mean-reverting bridge

In storage_extensions.py, SciPy generates 358 Sobol coordinates per path, which we transform into standard normal inputs using the inverse normal distribution. We draw the final log-price first, then fill midpoints using the exact Ornstein–Uhlenbeck conditional distribution below. Exponentiation gives daily spot prices.

For $l<t<r$, define $a=e^{-\kappa(t-l)}$, $b=e^{-\kappa(r-t)}$, $q_1=q_{t-l}$, $q_2=q_{r-t}$, $A=m+a(X_l-m)$ and $B=m+ab(X_l-m)$. Conditional covariance is bq_1 and endpoint variance is $b^2q_1+q_2$; Gaussian conditioning gives our adaptation:

$$X_t|X_l,X_r \sim N\left(A+\frac{bq_1}{b^2q_1+q_2}(X_r-B),\ \frac{q_1q_2}{b^2q_1+q_2}\right).$$

The construction preserves the exact daily joint law; covariance checks give errors below 5×10^{-16}. Every path then follows the same fitted operating rule, using only current spot and inventory.

4.3 - Results of the sampling comparison

We compare antithetic Monte Carlo, scrambled Sobol in date order, and Sobol with the mean-reverting bridge. All three value the same fitted strategy in the high-volatility case.

a) Measure error against the same strategy

We use 16 independent repetitions at 8,192 and 16,384 paths, retaining Sobol's power-of-two balance. The two Sobol constructions share scrambled points within each repetition. Comparing each estimate $\hat{V}_r$ with the strategy's independently integrated value, V_{ref}=EUR $4{,}505{,}786$, isolates sampling error from decision quality:

$$\text{RMSE}=\sqrt{\frac{1}{16}\sum_{r=1}^{16}\left(\hat{V}_r\text{-}V_{\text{ref}}\right)^2}, \qquad \text{standard error}\left(\overline{V}\right)=s_R/\sqrt{16}.$$

Root-mean-square error (RMSE) measures the typical pricing error across the 16 repetitions; s_R is their sample standard deviation. The standard error refers to the average of those estimates. For Sobol, uncertainty is measured across independent scrambles because points within one scramble are dependent.

b) Compare precision and computation time

Table 4.9. Sampling comparison: 16,384 paths per run and 16 independent repetitions.

Method	Mean error (EUR)	Run standard deviation (EUR)	Root mean square error (EUR)	Time per run (s)
Antithetic Monte Carlo	−4,134	8,929	9,583	2.32
Sobol: chronological	−555	3,631	3,559	2.48
Sobol: bridge	−248	2,540	2,472	2.43

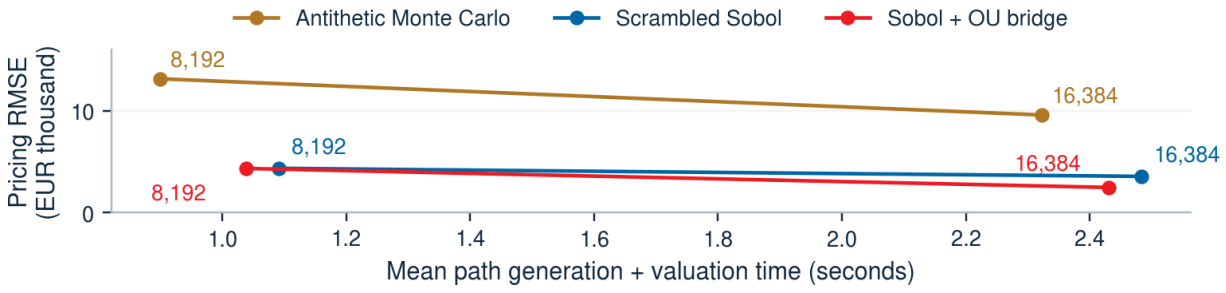


Figure 4.4. Pricing root-mean-square error versus path generation and valuation time. Lower points indicate greater precision; labels show paths per run. Four workers, one BLAS thread each; loading and fitting are excluded.

c) What the improvement means

At 16,384 paths, the bridge reduces root-mean-square error to EUR 2,472: 74% below antithetic Monte Carlo and 31% below chronological Sobol, with similar evaluation time. At 8,192 paths, the two Sobol methods nearly tie (EUR 4,351 and EUR 4,339), both below Monte Carlo's EUR 13,132.

We recommend scrambled Sobol with the mean-reverting bridge for evaluating this strategy at 16,384 paths. The operating rule and its expected cashflow are unchanged; we estimate that cashflow more precisely at comparable computational cost.

Practical conclusions and references

The project brings the methods developed in the first three exercises together in a practical gas storage implementation. We extended the option regression framework to handle inventory, daily trading choices and the terminal inventory requirement. Shared regressions and vectorized action comparisons keep this added complexity manageable. Every evaluated path satisfies the operating constraints, and the fitted decisions, valued independently by numerical integration, achieve values within 0.04% of the optimized reference in both volatility cases. We also chose to build an interactive visual tool to make the dynamic programming easier to understand and show how prices, future value and operating constraints translate into daily decisions.

We then tested where improvement was worthwhile. Halving inventory spacing to 1,250 MWh adds no optimized value in either case while requiring about 3.6 times the fitting time, supporting the original grid. The sampling study delivers a substantial improvement: at 16,384 paths in the high-volatility case, scrambled Sobol with a mean-reverting bridge reduces valuation root-mean-square error by 74% versus antithetic Monte Carlo and 31% versus chronological Sobol, with similar evaluation time. We can therefore value the same operating strategy much more precisely at a comparable computational cost.

The earlier exercises provide the foundation: agreement with published option prices, a European control reducing price-estimator variance by roughly three quarters, and a measured comparison of Euler and Milstein accuracy and cost. The larger gamma bump also provides supporting evidence for a practical way to improve sensitivity estimation. Together, these tools give us a modular gas storage implementation whose decisions we can explain, whose accuracy we can check independently, and whose improvements we can assess through numerical results.

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